

INTERMEDIATE-CARDINALITY STRATA AND DIRECT LIMITS IN LEAN: A FORMALIZATION STUDY AROUND THE CONTINUUM

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ABSTRACT. This paper draft presents a Lean 4 formalization study of an Intermediate-Cardinality Arithmetic Hypothesis (ICAH): a stratified view of the real continuum under the negation of the Continuum Hypothesis. The development is organized as two pillars. **Pillar A** proves the main theorem under the single hypothesis $\neg \text{CH}$: strata are elementary substructures of $\mathbb{R}$ in the first-order ring language, produced at every intermediate cardinality by Mathlib’s downward Löwenheim–Skolem theorem, so elementarity is free by construction. **Pillar B** gives a concrete algebraic realization — strata are relative algebraic closures of generated subfields, with native real-closed-field instances — conditional on one further hypothesis, the model completeness of real-closed fields (Tarski–Seidenberg), the single remaining Mathlib gap. The formalization proves, among other results: that $\mathbb{R}$ is a real-closed field; that for every cardinal $\aleph_0 \leq \kappa \leq 2^{\aleph_0}$ there is a real-closed subfield of $\mathbb{R}$ of cardinality exactly κ ; Tarski–Vaught elementarity theorems for direct limits of elementary chains, both relativized to $\mathbb{R}$ and ambient-free (the canonical maps into the direct limit are elementary); the cardinality identity $\#F_\omega = \sup_n \#C_n$ for such direct limits, together with a closure theorem (countable chains of intermediate strata never exhaust $\mathbb{R}$) and a sharpness theorem (the least length of an exhausting family of intermediate strata is exactly $\text{cof}(2^{\aleph_0})$). The project declares zero axioms: the two named hypotheses are threaded explicitly through the theorem statements, and every flagship result is machine-audited in the build to depend only on the Lean kernel axioms.

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1. INTRODUCTION

The Continuum Hypothesis (CH) asserts that there is no cardinal strictly between the cardinality of the natural numbers and the cardinality of the continuum. Consequently, any mathematical program that treats intermediate cardinalities inside $\mathbb{R}$ must either work relative to $\neg \text{CH}$, change the meaning of size, or explicitly track which assumptions are external to ZFC.

The Lean development studied here chooses the first route. It introduces a named proposition, `ICAH.NotCH : Prop`, defined as `continuum ≠ aleph 1`, and threads it through the development as an explicit hypothesis. Around this hypothesis, the project builds a formal architecture for strata of the real continuum, size-aware field structures, subfield refinements, elementary substructures, elementary chains, and direct limits.

1.1. What “ICAH” denotes.

The name requires precision. Despite the word “Hypothesis”, ICAH is **not** an axiom candidate in the sense of Martin’s Axiom or PFA: it is not a new assumption whose consistency strength is at issue. ICAH denotes a **theorem schema under $\neg \text{CH}$** — a conjunction of structural claims about intermediate-cardinality strata of $\mathbb{R}$, each provable (in ZFC, formalized in Lean/Mathlib) once $\neg \text{CH}$ is

granted. The formal content is the pair of Lean propositions `ICAHElementary` and `ICAHStatement` displayed below, together with their proofs.

1.2. Main theorem.

The development is organized as two pillars that realize the same informal picture at different levels of concreteness.

Theorem. (Theorem 1 — Pillar A, semantic form; Lean: `icahElementary`.) Assume $\neg \text{CH}$ (i.e. `NotCH`). Then, writing $\mathcal{L}$ for the first-order ring language and “stratum” for an elementary substructure $S \prec \mathbb{R}$ in $\mathcal{L}$:

1. **(band)** there exists a cardinal κ with $\aleph_0 < \kappa < 2^{\aleph_0}$;
2. **(strata at every level)** for every cardinal κ with $\aleph_0 < \kappa < 2^{\aleph_0}$ there is an elementary substructure $S \prec \mathbb{R}$ with $\#S = \kappa$ (this clause is ZFC-pure: downward Löwenheim–Skolem);
3. **(chains)** there is an $\mathbb{N}$ -indexed elementary chain of intermediate-size strata whose direct limit is elementarily equivalent to $\mathbb{R}$;
4. **(closure; ZFC-pure)** for every $\mathbb{N}$ -indexed elementary chain whose levels have size $< 2^{\aleph_0}$, the direct limit has size $< 2^{\aleph_0}$ — countable chains never escape the hierarchy;
5. **(cofinality)** every real number lies in some intermediate-size elementary substructure of $\mathbb{R}$.

The Lean proof depends only on the kernel axioms `propext`, `Classical.choice`, `Quot.sound`; this is machine-checked by `#guard_msgs` in the build.

Theorem. (Theorem 2 — Pillar B, algebraic realization; Lean: `icahTheorem`.) Assume $\neg \text{CH}$ and additionally `RCFModelComplete` (every real-closed subfield of $\mathbb{R}$ is an elementary substructure in the ring language — model completeness of RCF, the single remaining Mathlib gap). Then the four clauses of `ICAHStatement` hold:

1. **(M1)** for every ordinal n there is a stratum R with index n and $\aleph_0 < \#R < 2^{\aleph_0}$;
2. **(M3)** every stratum carries a size-aware ordered-field structure with matching carrier and cardinal;
3. **(M5)** there is a chain of strata — concretely, real-closed subfields of $\mathbb{R}$ given as relative algebraic closures — whose direct limit is elementarily equivalent to $\mathbb{R}$;
4. **(M6)** there is a `ℵ`.ord-indexed monotone family of intermediate-size subfields of $\mathbb{R}$ whose union has cardinality $2^{\aleph_0}$.

The duality between the pillars is a genuine trade-off, made explicit throughout the paper: Pillar A gets elementarity for free (the strata are Skolem hulls, with no concrete algebraic description beyond being subfields), while Pillar B has concrete, named strata (relative algebraic closures, with native `IsRealClosed` instances) but must purchase elementarity through Tarski–Seidenberg.

Contribution. The main contribution is not new mathematics — to a model theorist, the existence of real-closed subfields of every intermediate cardinality

is a corollary of downward Löwenheim–Skolem plus Tarski, and the chain results are textbook Tarski–Vaught. The contribution is the formalization architecture and the assumption cartography: a Lean-readable decomposition of a continuum-stratification statement into proved lemmas, explicitly threaded hypotheses, and Mathlib-facing gaps, with the dependency set of every flagship theorem machine-audited in the build. The development declares **zero** axioms; the main theorem (Pillar A) holds under $\neg \text{CH}$ alone.

The project should be positioned for a mathematical audience as a formalization paper with three layers:

1. a set-theoretic layer, where the hypothesis $\neg \text{CH}$ produces intermediate cardinal witnesses;
2. a model-theoretic/algebraic layer, where elementary substrata of every intermediate cardinality are produced by downward Löwenheim–Skolem (Pillar A), and real-closed subfields of every infinite cardinality up to the continuum are constructed and assembled into elementary chains (Pillar B);
3. a proof-engineering layer, where the one remaining Mathlib dependency is exposed as a named hypothesis, machine-audited in the build, and can be attacked as an independent Mathlib contribution.

The present paper draft is organized as follows. Section 2 recalls the mathematical background. Section 3 describes the Lean architecture. Section 4 isolates the results that are already proved. Section 5 gives the hypothesis inventory, the per-theorem audit, and the Mathlib roadmap. Section 6 discusses related work, including the formalizations of the independence of CH and the certified quantifier-elimination procedures for real-closed fields in other proof assistants. Section 7 briefly notes analogues in valued and p-adic field theory. Section 8 concludes with reproducibility data and an engagement plan.

1.3. Guiding problem.

Definition. An *intermediate-cardinality stratum* is, informally, a subset $S \subseteq \mathbb{R}$ equipped with a cardinal κ such that

$$\aleph_0 < \kappa < 2^{\aleph_0}$$

and a proof that the subtype determined by S has cardinality κ .

Definition. A *size-aware field* is a bundled field-like object carrying its underlying type, a designated cardinal, and the typeclass instances needed to treat the carrier as an ordered ring or field in Lean.

The central formal question is:

Question. Can one build, under explicit assumptions, a hierarchy of intermediate-size strata whose carriers support enough algebra and model theory to form elementary chains, and how long must a family of such strata be to exhaust $\mathbb{R}$?

The current Lean code answers this question completely. Semantically (Pillar A), every clause is proved under $\neg \text{CH}$ alone. Algebraically (Pillar B), every clause is proved once the model completeness of real-closed fields is granted. And the exhaustion question has a sharp answer: the least length of a family of intermediate strata covering $\mathbb{R}$ is exactly $\text{cof}(2^{\aleph_0})$.

2. BACKGROUND

2.1. The continuum and the role of $\neg \text{CH}$.

Let $c = 2^{\aleph_0}$ denote the cardinality of the continuum. The Continuum Hypothesis states that there is no cardinal κ satisfying $\aleph_0 < \kappa < c$. Since Gödel and Cohen, the standard mathematical background is that CH is independent of ZFC, assuming ZFC is consistent [1, 2].

The Lean project therefore does not attempt to prove $\neg \text{CH}$ in ZFC. Instead, it defines $\neg \text{CH}$ as a named proposition (`ICAH.NotCH`) and threads it as an explicit hypothesis through every theorem that needs it. The environment contains no project axioms; the per-theorem `#print axioms` audit reports only the Lean kernel axioms.

Proposition. Under the hypothesis `ICAH.NotCH`, the cardinal $\aleph_1$ witnesses the existence of an intermediate cardinal:

$$\exists \kappa, \aleph_0 < \kappa \wedge \kappa < 2^{\aleph_0}.$$

The Lean proof uses the standard inequality $\aleph_0 < \aleph_1$, the theorem $\aleph_1 \leq c$, and converts the inequality plus non-equality $c \neq \aleph_1$ into the strict inequality $\aleph_1 < c$.

2.2. Real closed fields and definability.

Real closed fields are the algebraic and model-theoretic setting closest to the ordered real field. They admit quantifier-elimination results and provide the natural language in which order, addition, multiplication, and definability can be studied together [3, 4].

ICAH uses this background in a restrained way. The code does not merely assume every bare subset of $\mathbb{R}$ is a field. Instead, it introduces `SubfieldStratum`, which requires the carrier set to be the underlying set of a `Subfield RR`. This is the right formal move: closure under field operations is obtained from the subfield structure rather than reconstructed ad hoc.

2.3. Elementary substructures and downward Löwenheim–Skolem.

An elementary substructure $S \prec M$ satisfies the same first-order formulas as M at parameters from S , by definition. The downward Löwenheim–Skolem theorem produces such substructures at every prescribed infinite cardinality between the language size and $\#M$ [5]. Mathlib formalizes this as `exists_elementarySubstructure_card_eq` (via Skolem functions), and Pillar A of the present development is essentially a specialization of that theorem to $\mathbb{R}$ in the ring language: elementarity of the strata is free by construction.

2.4. Elementary chains and direct limits.

An elementary chain is a sequence of structures connected by elementary embeddings. The Tarski–Vaught elementary chain theorem asserts that the union (or directed limit) of such a chain is an elementary extension of each level [5, 6].

The Lean project packages this through `ElemChain`, where each level carries an `LOR`-structure and each successor map is an elementary embedding. The direct limit is then implemented using Mathlib’s `Language.DirectLimit` API. The project proves the chain theorem **relativized to the ambient model $\mathbb{R}$** (`tarskiVaughtDirectLimit`); an earlier draft misattributed this result to Łoś, whose theorem concerns ultraproducts — the misnomer has been corrected throughout.

3. FORMAL ARCHITECTURE

This section records the paper-level interpretation of the Lean objects. The goal is to make the repository legible to mathematicians without forcing them to read Lean code first.

The development is organized as two pillars sharing the cardinal scaffolding:

1. **Pillar A** (`ICAH/ElementaryStrata.lean`): strata are elementary substructures of $\mathbb{R}$ in the ring language, produced by downward Löwenheim–Skolem. Elementarity is definitional; the main theorem `icahElementary` needs only `NotCH`.
2. **Pillar B** (`ICAH/FieldOnStratum.lean` and downstream): strata are concrete real-closed subfields (relative algebraic closures of generated subfields). Elementarity is purchased through the `RCFModelComplete` hypothesis.

3.1. Core objects.

Definition. `Stratum` is the basic continuum-layer object. It consists of an ordinal index, a set $S \subseteq \mathbb{R}$, a cardinal κ , a witness that the subtype determined by S has cardinality κ , and bounds $\aleph_0 < \kappa < c$.

Definition. `SizeAwareField` is a bundled structure containing a carrier type, a designated cardinal, and the instances `[Field K]`, `[LinearOrder K]`, and `[IsStrictOrderedRing K]`. This avoids relying on deprecated or overly strong ordered-field classes while still exposing enough structure for model-theoretic use.

Definition. `SubfieldStratum` refines `Stratum` by adding a subfield `subfield : Subfield RR` and a proof that the stratum set is exactly the underlying set of that subfield.

The role of `SubfieldStratum` is central. A raw set of real numbers is not automatically closed under addition, multiplication, negation, and inverses. A subfield is. Thus, the field part of the theory is moved from a fragile closure proof obligation into a stable algebraic structure.

Definition. `RCSubfieldStratum` refines `SubfieldStratum` once more by requiring the subfield to be **real closed**. This is the correct hypothesis for model theory:

a general subfield of $\mathbb{R}$ (such as $\mathbb{Q}$) is not elementarily embedded in $\mathbb{R}$, while a real-closed subfield is, by model completeness of the theory of real-closed fields.

Definition. In Pillar A, a stratum is an `LOR.ElementarySubstructure` $\mathbb{R}$ — Mathlib’s bundled elementary substructure. The theorem `exists_elementary_substratum` produces one of every infinite cardinality $\kappa \leq 2^{\aleph_0}$ (optionally containing a prescribed set of size $\leq \kappa$, via `exists_elementary_substratum_extending`), by instantiating Mathlib’s downward Löwenheim–Skolem theorem `exists_elementarySubstructure_card_eq` at $M = \mathbb{R}$. The bridge lemma `elemSubstratumSubfield` shows every such stratum is the carrier of a subfield of $\mathbb{R}$: closure under ring operations is the substructure property, and closure under inverses is one formula transfer ($\exists y, x \cdot y = 1$) along elementarity.

Construction. `subfieldToSAF` converts a subfield of $\mathbb{R}$, together with a cardinality witness, into a `SizeAwareField`. The key design choice is to use subtype inheritance for the linear order and Mathlib’s subfield ordered-ring instance for the strict ordered ring structure.

Construction. `relAlgebraic` K_0 is the relative algebraic closure of a subfield $K_0 \subseteq \mathbb{R}$ inside $\mathbb{R}$. It is proved real closed via a root-closure criterion, and its cardinality is bounded by the maximum of the cardinality of K_0 and $\aleph_0$. This is the engine behind `exists_rc_subfield`, which produces a real-closed subfield of any prescribed infinite cardinality up to the continuum.

3.2. Definability layer.

The language must be stated precisely: `LOR` is an abbreviation for Mathlib’s `Language.ring` — the first-order language with function symbols $+$, $\cdot$, $-$, 0 , 1 and `no` relation symbols. It is not a language of ordered rings: there is no order symbol. This costs nothing for the intended models, because in a real-closed field the order is definable from the ring structure ($x \leq y$ iff $y - x$ is a square), but it matters for honesty about the definability results below.

In the current development, the graphs of addition and multiplication on $\mathbb{R}$ are proved definable by constructing bounded formulas and transporting realization through Mathlib’s language-homomorphism machinery (`graphDefinable_add`, `graphDefinable_mul`).

Remark. Since `LOR` contains the ring symbols, the graphs of $+$ and $\cdot$ are **atomic** formulas, and the mathematical content of these definability lemmas is nil. They are presented honestly as what they are: an API stress test of Mathlib’s realization machinery — bounded formulas, `Sum`-variable bookkeeping, and the `CompatibleRing` transfer — which is exactly the plumbing later needed for the elementarity transfers in both pillars. Had the language been a pure order language, addition would **not** be definable; the precise choice of `LOR` is therefore load-bearing.

3.3. Elementary chains.

Definition. `ElemChain` packages a sequence `obj : ℕ → Type*` of first-order structures and elementary embeddings `obj n ↪e obj (n+1)`.

From these successor maps, the project defines two related systems:

1. `embLE`, which composes successor elementary embeddings and preserves elementarity;
2. `sysEmb`, which uses Mathlib’s directed-system API to build the underlying embedding system.

The lemma `embLE_eq_sysEmb` proves that these two constructions agree as functions. This is a small but important bridge: the proof-relevant elementary embedding API and the directed-colimit API are not automatically the same object.

3.4. Direct limit.

Definition. For an elementary chain `C`, the direct limit `DirectLim C` is defined as `Language.DirectLimit C.obj (...)`, using the underlying directed system of embeddings.

The direct limit is the formal version of the intended limit field F_ω . The project proves the cardinality identity $\#F_\omega = \sup_n \#C_n$ (`directLimit_card_eq_iSup`), the closure theorem that countable chains of intermediate strata stay intermediate (`directLimit_card_lt_continuum`, via König), and two elementarity theorems: the relativized version over $\mathbb{R}$ (`tarskiVaughtDirectLimit`, via the Tarski–Vaught test) and the ambient-free version (`ofLevelElem`: the canonical maps into the direct limit are elementary, by induction on bounded formulas). Both elementarity statements were previously exposed as gaps and are now theorems.

3.5. Cofinal family at the continuum, and optimality.

The $\mathbb{N}$ -indexed chain cannot, by itself, exhaust $\mathbb{R}$: König’s theorem gives $\text{cof}(c) > \aleph_0$, so a countable increasing union of sets of size $< c$ has size $< c$. The module `ICAH/CofinalFamily.lean` therefore introduces a `ℵ`-indexed monotone family of intermediate-size real-closed subfields whose union is all of $\mathbb{R}$ (`exists_cofinal_rc_family`). The top-level statement `ICAHStatement` uses this family for its limit-size clause, which keeps the formalized claim non-vacuous.

The module also proves that $\text{cof}(c)$ is the **exact** threshold: any family of subsets of $\mathbb{R}$ of size $< c$ covering $\mathbb{R}$ has at least $\text{cof}(c)$ members (`cofinal_family_length_lower_bound`), and a monotone real-closed covering family of length exactly $\text{cof}(c)$ exists (`exists_cofinal_rc_family_cof_length`, by composing the `ℵ`-indexed family with a fundamental sequence). See Section 4.

4. PROVED RESULTS

This section separates fully proved Lean results from hypotheses. Section 8 contains the complete declaration-to-paper mapping table; Section 5 records, for each result, which of the two named hypotheses (if any) it consumes.

4.1. Intermediate cardinal witness.

Theorem. `exists_intermediate_cardinal`: assuming the hypothesis `NotCH`, there exists a cardinal κ such that $\aleph_0 < \kappa < c$.

Remark. The witness is $\aleph_1$. The proof uses `aleph0_lt_aleph_one`, `aleph_one_le_continuum`, and turns non-equality into strict inequality (`aleph_one_lt_continuum_of_notCH`).

4.2. Elementary substrata via downward Löwenheim–Skolem (Pillar A).

Theorem. `exists_elementary_substratum_extending` (ZFC-pure): for every cardinal $\aleph_0 \leq \kappa \leq c$ and every set $s \subseteq \mathbb{R}$ with $\#s \leq \kappa$, there is an elementary substructure $S \prec \mathbb{R}$ in the ring language with $s \subseteq S$ and $\#S = \kappa$.

Remark. This is Mathlib’s downward Löwenheim–Skolem theorem `exists_elementarySubstructure_card_eq` instantiated at $M = \mathbb{R}$; the language has five symbols (`card_ring`), so the language-size condition is absorbed by $\aleph_0 \leq \kappa$. Elementarity of the strata is free — no Tarski–Seidenberg. This single theorem powers every clause of `icahElementary` (Theorem 1).

Theorem. `elemSubstratumSubfield`: the carrier of any elementary substructure of $\mathbb{R}$ in the ring language is a subfield of $\mathbb{R}$.

Remark. Closure under $+, \cdot, -, 0, 1$ is the substructure property; closure under inverses is a transfer of the formula $\exists y, x \cdot y = 1$ along elementarity. This is the satisfaction-to-structure bridge that gives Pillar A strata an algebraic identity, partially compensating for their abstract Skolem-hull origin.

4.3. Concrete synthetic stratum.

Construction. `syntheticStratum`: under `NotCH`, materializes an intermediate-cardinality subset of $\mathbb{R}$ by extracting a concrete set $S \subseteq \mathbb{R}$ with cardinality $\aleph_1$ from a cardinal existence theorem.

This is the first bridge from a cardinal statement to an inhabited Lean structure. It turns the abstract existence of an intermediate cardinal into a concrete `Stratum` object.

4.4. Subfield strata and size-aware fields.

Theorem. `fieldOnSubfieldStratum`: every `SubfieldStratum` yields a `SizeAwareField` whose carrier and cardinal match the underlying stratum.

The technical point is that the carrier equality is treated as a propositional equality of subtypes, not merely a bijection. This is precisely what the `SizeAwareField.carrier` field requires.

Lemma. `fieldOnStratum`: every `Stratum` (not only subfield strata) admits a `SizeAwareField` with matching carrier and cardinal. Formerly an axiom.

Remark. This is deliberately stated as a lemma, not a theorem: it is a **packaging** result. The proof transports the field structure of a real-closed subfield of the right cardinality (from `exists_rc_subfield`, below) across a cardinality `Equiv`, using `Equiv.field` and `LinearOrder.lift'` — so the resulting operations on the carrier are an arbitrary transport and bear no relation to the ambient operations of $\mathbb{R}$. The mathematically meaningful field structures live on `SubfieldStratum` and `RCSubfieldStratum`, whose operations **are** those of $\mathbb{R}$.

4.5. Real-closedness of $\mathbb{R}$ and of relative algebraic closures.

Theorem. `Real.isRealClosed`: $\mathbb{R}$ is a real-closed field. Formerly an axiom.

Remark. The proof uses `IsRealClosed.of_linearOrderedField`: squares are handled by `Real.sqrt`, and odd-degree polynomials acquire roots by the intermediate value theorem combined with the leading-coefficient asymptotics `Polynomial.tendsto_atTop/atBot`.

Theorem. `isRealClosed_of_forall_root` and `relAlgebraic_isRealClosed`: a subfield of $\mathbb{R}$ closed under taking real roots of its polynomials is real closed; in particular the relative algebraic closure `relAlgebraic K₀` of any subfield $K_0 \subseteq \mathbb{R}$ is real closed.

Theorem. `exists_rc_subfield` (ZFC-pure): for every cardinal κ with $\aleph_0 \leq \kappa \leq c$, there exists a real-closed subfield of $\mathbb{R}$ of cardinality exactly κ .

Remark. This proves (and strengthens) the former `subfieldStratumExists` axiom. The subfield is the relative algebraic closure of the subfield generated by a set of size κ ; the cardinality is pinned by `Algebra.IsAlgebraic.cardinalMk_le_max` for the upper bound and injectivity of the generators for the lower bound. It is the concrete (Pillar B) counterpart of `exists_elementary_substratum`: an explicit algebraic description, at the price that elementarity over $\mathbb{R}$ is no longer free.

4.6. Real algebraic numbers as a base field.

Theorem. `algReal_card_le_aleph0`: the real algebraic numbers, represented as `algebraicClosure QQ RR` inside $\mathbb{R}$, have cardinality at most $\aleph_0$.

Together with the lower bound for infinite types, this supports the concrete base object `algRealSAF`, a countable size-aware field outside the intermediate-cardinality hierarchy.

4.7. Definability of addition and multiplication.

Lemma. `graphDefinable_add` and `graphDefinable_mul`: the graphs of addition and multiplication on $\mathbb{R}$ are first-order definable in the ring language `LOR` with parameters.

Remark. Stated honestly: since `LOR` contains the ring function symbols, these graphs are atomic formulas and the lemmas carry no mathematical content. Their value is as an API stress test — building explicit bounded formulas and verifying realization through several layers of Mathlib’s model-theory machinery (Sum-variable management, `CompatibleRing` transfer) — which is the same plumbing used by the elementarity transfers elsewhere in the development. The contrast case is real: in the pure order structure $(\mathbb{R}, <)$ addition is not definable, so the choice of language is load-bearing.

4.8. Elementarity-preserving composition.

Theorem. `embLE`: for any $m \leq n$, the chain induces an elementary embedding from level m to level n .

Theorem. `embLE_eq_sysEmb`: the elementarity-preserving embedding built by recursive composition agrees as a function with the directed-system embedding used by `Language.DirectLimit`.

4.9. Tarski–Vaught theorem for direct limits.

Theorem. `ElemChain.tarskiVaughtDirectLimit`: if every level of an elementary chain embeds elementarily and compatibly into $\mathbb{R}$, then the direct limit is elementarily equivalent to $\mathbb{R}$. Formerly an axiom.

Remark. The underlying map is built by `Language.DirectLimit.lift`; elementarity is established by the Tarski–Vaught test: a witness of an existential in $\mathbb{R}$ over parameters from the limit is pulled back to a finite level via the compatibility of the embeddings. **Naming note:** an earlier draft called this result “Łoś-style”; that was a misattribution (Łoś’s theorem concerns ultraproducts; the relevant classical result is the Tarski–Vaught elementary chain theorem, here relativized to the ambient model $\mathbb{R}$).

Theorem. `ElemChain.ofLevelElem` and `realize_ofLevel_iff` (ZFC-pure; **ambient-free** Tarski–Vaught chain theorem): for any elementary chain, the canonical embedding of each level into the direct limit is itself **elementary** — no ambient model required. Consequently the direct limit is elementarily equivalent to every level (`directLim_elementarilyEquivalent`).

Remark. Unlike the relativized version, this cannot be obtained from Mathlib’s Tarski–Vaught test (pulling realization back through `DirectLimit.of` for arbitrary formulas is precisely what is being proved). The proof is a direct induction on `BoundedFormula`; in the `all` case, a direct-limit witness is pulled down to a common level via directedness and `DirectLimit.of_f`, while the universally quantified hypothesis is pushed up through the elementary transition embed-

dings. Mathlib’s `DirectLimit` file currently has no elementarity results at all; this theorem is the form Mathlib should receive.

4.10. Cardinality of the direct limit: identity, closure, and sharpness.

An earlier draft stated `directLimit_card`: if every level has cardinality below the continuum **and** the supremum of the level cardinalities is the continuum, **then** the direct limit has cardinality continuum. Those hypotheses are jointly unsatisfiable in ZFC — König’s theorem gives $\text{cof}(c) > \aleph_0$, so a countable supremum of cardinals $< c$ stays $< c$ — making the theorem true and empty. It has been deleted and replaced by three non-vacuous results.

Theorem. `directLimit_card_eq_iSup` (ZFC-pure): for an $\mathbb{N}$ -indexed chain with infinite supremum of level cardinalities, $\#(\text{DirectLim } C) = \sup_n \#C_n$.

The proof has two halves. The upper bound exhibits the direct limit as a quotient of the sigma type $\sum_{n:\mathbb{N}} C_n$, bounding its cardinality by $\aleph_0 \cdot \sup_n \#C_n = \sup_n \#C_n$. The lower bound uses injectivity of the canonical maps from each level into the direct limit and then takes the supremum over levels.

Theorem. `directLimit_card_lt_continuum` and `directLimit_intermediate` (ZFC-pure; closure): if every level of an $\mathbb{N}$ -indexed chain has cardinality $< c$, so does the direct limit; if moreover the first level is uncountable, the limit is again of intermediate size. Countable elementary chains never escape the stratum hierarchy.

Theorem. `cofinal_family_length_lower_bound` (ZFC-pure; sharpness, lower bound): any family $\{S_i\}_{i \in \iota}$ of subsets of $\mathbb{R}$ with $\#S_i < c$ for all i and $\bigcup_i S_i = \mathbb{R}$ satisfies $\#\iota \geq \text{cof}(c)$.

Theorem. `exists_cofinal_rc_family_cof_length` (under `NotCH`; sharpness, upper bound): there is a monotone family of intermediate-size real-closed subfields of $\mathbb{R}$, indexed by the ordinal $(\text{cof}(c)).\text{ord}$, covering all of $\mathbb{R}$.

Remark. Together, the last two results say: **the least length of an exhausting family of intermediate strata is exactly $\text{cof}(c)$** — a genuine sharpness statement in the language of cardinal invariants. The upper bound composes the `ℵ.ord`-indexed family below with a fundamental sequence (`Ordinal.exists_isFundamentalSeq`). The König obstruction that made the original formulation vacuous is thereby converted into the optimality half of a theorem.

4.11. Limit size via the cofinal family.

Theorem. `exists_cofinal_rc_family` and `cofinal_family_limit_size` (under `NotCH`): there is a monotone, `ℵ.ord`-indexed family of real-closed subfields of

$\mathbb{R}$, each of intermediate cardinality, whose union is all of $\mathbb{R}$ and hence has cardinality c .

Remark. The shift from an $\mathbb{N}$ -index to an uncountable-cofinality index is forced by König's theorem, as quantified precisely by the sharpness results above. This corrected formulation is what `ICAHStatement` asserts for the limit-size clause.

Contribution. For the paper, `exists_elementary_substratum`, `tarskiVaughtDirectLimit`, `directLimit_card_eq_iSup`, the $\text{cof}(c)$ sharpness pair, and `exists_rc_subfield` are the strongest fully proved results to foreground. Each is independent of the grand interpretation of ICAH, and all but the `NotCH`-consuming ones are ZFC-pure — which is precisely their selling point as Mathlib upstreaming candidates.

5. HYPOTHESIS INVENTORY, PER-THEOREM AUDIT, AND MATHLIB ROADMAP

A central strength of the Lean development is that it does not hide its assumptions. After the June 2026 refactor, the project declares **zero** axioms: the two named assumptions are ordinary `Props` threaded through the development as explicit hypotheses, so the assumption set of every theorem is visible in its type signature. The kernel-level audit is enforced in-source by `#guard_msgs` in `#print axioms` blocks and checked in CI:

```
'ICAH.icaHElementary' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'ICAH.icaHTheorem' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

Any drift in these sets — including a hidden `sorryAx` — fails the build. With hypotheses in signatures rather than axioms in the environment, `#guard_msgs` is demoted from safety mechanism to regression test, which is the right division of labor.

5.1. External mathematical hypothesis.

Assumption. `ICAH.NotCH` : `Prop`, defined as `continuum ≠ aleph 1`: the negation of the Continuum Hypothesis.

This is not a Mathlib gap. It is the intended set-theoretic regime of the project: the theory is developed relative to $\neg \text{CH}$, which is consistent with ZFC by Cohen’s forcing argument [2] — itself formalized in Lean by the Flypitch project [7] (see Section 6).

5.2. The single remaining Mathlib gap.

Mathlib gap. `ICAH.RCFModelComplete` : `Prop`: model completeness of the theory of real-closed fields (a consequence of Tarski–Seidenberg quantifier elimination). Concretely: the inclusion of a real-closed subfield of $\mathbb{R}$ into $\mathbb{R}$ is an elementary embedding in the ring language (in which the order of a real-closed field is definable via squares).

This is a true classical theorem whose first-order formalization (quantifier elimination or model completeness for RCF inside Mathlib’s `ModelTheory` framework) is absent from Mathlib. It is consumed only by Pillar B (`icaHTheorem`); Pillar A (`icaHElementary`) avoids it entirely via downward Löwenheim–Skolem.

Remark. Why this gap is harder than the ACF precedent. Mathlib already contains the deep-embedded model theory of algebraically closed fields: the theory `ACF p` over the ring language, its completeness for p prime or zero (`ACF_isComplete`), and the Lefschetz principle. That suggests a template — but completeness of ACF comes cheap via uncountable categoricity and the Łoś–Vaught test, a route that is **closed** for RCF: the theory of real-closed fields is unstable (it defines a linear order) and has the maximum number of models in every uncountable cardinality. Discharging `RCFModelComplete` therefore requires QE-grade work — e.g. Robinson’s model-completeness test with sign-change/

root-counting embedding arguments inside the deep embedding — substantially more than porting the ACF files.

Prior art calibrates the effort: the `math-comp/real-closed` library in Coq/Rocq contains a certified quantifier-elimination procedure for RCF with the decision procedure `rcf_sat` and its correctness proof (Cohen–Mahboubi [8]), and HOL Light has McLaughlin–Harrison’s proof-producing decision procedure for RCF [9]. Neither transfers directly to Mathlib’s `FirstOrder` framework, but Cohen–Mahboubi is the closest blueprint. (This also discharges the earlier draft’s TODO asking for a situated estimate of the gap.)

5.3. Per-theorem audit.

Because the hypotheses are explicit, the audit can be given per theorem. The classification below is the contractual content of the development:

1. **ZFC-pure** (kernel axioms only, no project hypotheses): `exists_elementary_substratum`, `exists_elementary_substratum_extending`, `elemSubstratumSubfield`, `exists_rc_subfield`, `relAlgebraic_isRealClosed`, `isRealClosed_of_forall_root`, `Real.isRealClosed`, `tarskiVaughtDirectLimit`, `ofLevelElem` (the ambient-free chain theorem), `directLimit_card_eq_iSup`, `directLimit_card_lt_continuum`, `directLimit_intermediate`, `cofinal_family_length_lower_bound`, `fieldOnStratum`, `fieldOnSubfieldStratum`, `algReal_card_le_aleph0`.
2. **Under NotCH**: `exists_intermediate_cardinal`, `syntheticStratum`, `subfieldStratumExists`, `intermediateRCSubfieldStratum`, `exists_cofinal_rc_family`, `exists_cofinal_rc_family_cof_length`, `cofinal_family_limit_size`, and the Pillar A main theorem `icahElementary`.
3. **Under NotCH and RCFModelComplete**: the Pillar B main theorem `icahTheorem` (the model-completeness hypothesis enters only through the elementary-chain clause M5).

The ZFC-pure results need neither hypothesis — that property is precisely their upstream selling point.

5.4. Resolved former axioms.

The earlier inventory was discharged in two stages. First, five axioms became theorems:

Theorem. `ICAH.Real.isRealClosed` — **proved** (`ICAH/RealClosed.lean`): $\mathbb{R}$ is real closed, via `IsRealClosed.of_linearOrderedField`, `Real.sqrt` for squares, and the intermediate value theorem for odd-degree polynomials.

Theorem. `ICAH.fieldOnStratum` — **proved** (`ICAH/FieldOnStratum.lean`): every stratum admits a size-aware field, by Equiv-transport of the field structure of a real-closed subfield of matching cardinality (a packaging lemma; see Section 4).

Theorem. `ICAH.subfieldStratumExists` — **proved and strengthened** (`exists_rc_subfield`): for every $\aleph_0 \leq \kappa \leq c$ there is a **real-closed** subfield of $\mathbb{R}$ of cardinality exactly κ , obtained as the relative algebraic closure of a generated subfield.

Theorem. `ICAH.ElemChain.losDirectLimit` — **proved and renamed** (`tarskiVaughtDirectLimit`, `ICAH/ElementaryChain.lean`): the direct limit of an elementary chain compatibly embedded in $\mathbb{R}$ is elementarily equivalent to $\mathbb{R}$, via `Language.DirectLimit.lift` and the Tarski–Vaught test.

Second, the two remaining axioms (`not_CH`, `rcfModelComplete`) were converted from axioms into the explicit hypotheses `NotCH` and `RCFModelComplete`, emptying the axiom inventory entirely.

Remark. Two further axioms of the earlier draft, `subfieldIsRealClosed` and `subfieldStratumElemEmb`, were **deleted as false in the stated generality**: $\mathbb{Q}$ is a subfield of $\mathbb{R}$ that is neither real closed nor elementarily embedded (the sentence $\exists x, x^2 = 2$ distinguishes $\mathbb{Q}$ from $\mathbb{R}$). They are replaced by the sound refinement `RCSubfieldStratum` (a stratum whose carrier is a real-closed subfield), for which elementarity follows from `RCFModelComplete`. This self-correction record — two former assumptions identified as false, with counterexample, and repaired by an honest weakening — is, we believe, the most scientifically credible feature of the methodology, and the reason the audit infrastructure exists.

5.5. The honest limit construction.

The original $\aleph$ -indexed chain formulation of the limit-size milestone is vacuous: König’s theorem gives $\text{cof}(c) > \aleph_0$, so no countable chain of intermediate-size strata can union to $\mathbb{R}$. The same disease affected the earlier cardinality theorem `directLimit_card`, whose hypotheses were jointly unsatisfiable; it has been replaced by the identity/closure/sharpness triple of Section 4. The corrected limit-size statement (`ICAH/CofinalFamily.lean`) indexes the family by the ordinal `ℵ.ord`:

Theorem. `exists_cofinal_rc_family`: there is a monotone family of real-closed subfields of $\mathbb{R}$, indexed by `ℵ.ord`, each of intermediate cardinality, whose union is all of $\mathbb{R}$; and `cofinal_family_limit_size`: the union has cardinality c . Moreover the index length can be improved to the optimum $\text{cof}(c)$ (`exists_cofinal_rc_family_cof_length`), and no shorter family exists (`cofinal_family_length_lower_bound`).

5.6. Recommended order of attack.

1. Post the minimized in-framework statement of `RCFModelComplete` on the Lean Zulip model-theory stream (resolving possible overlap with in-progress ordered-field/real-closure work) before any PR.
2. PR 1 (small, ZFC-pure, attractive): the DLS-based elementary-substrata existence for $\mathbb{R}$ plus the direct-limit cardinality identity `directLimit_card_eq_iSup`.

3. PR 2: the ambient-free Tarski–Vaught chain theorem for `Language.DirectLimit` (proved here as `ofLevelElem / realize_ofLevel_iff` for $\mathbb{N}$ -indexed chains in the ring language; the upstream version should generalize the index order and language — the formula induction carries over verbatim, with the relation case no longer vacuous).
4. PR 3 (after dedup check): `Real.isRealClosed`, the root-closure criterion `isRealClosed_of_forall_root`, and `relAlgebraic_isRealClosed`.
5. Long-term: formalize quantifier elimination / model completeness for RCF in Mathlib’s `ModelTheory` framework, modeled on the ACF files but via Robinson’s test, discharging `RCFModelComplete` — the final gap.

6. RELATED WORK

6.1. Formalizations of the independence of CH.

The set-theoretic background is classical: Gödel proved relative consistency of CH with ZFC, and Cohen proved the independence direction using forcing [1, 2]. Crucially for the present paper, this metatheory has itself been formalized. The **Flypitch** project of Han and van Doorn gave a formal proof of the independence of the Continuum Hypothesis in Lean, with both consistency directions obtained by forcing with Boolean-valued models — Cohen forcing for $\neg$ CH and collapse forcing for CH [7, 10]. Independently, Gunther, Pagano, Sánchez Terraf, and Steinberg formalized the independence of CH in Isabelle/ZF [11].

The contrast sharpens the positioning of ICAH in one sentence: Flypitch and the Isabelle/ZF line formalize the **metatheory** (unprovability, via forcing models), while ICAH works **object-theoretically inside** the $\neg$ CH regime, building structure on intermediate cardinalities rather than building models of set theory. The present work contributes nothing to the independence problem itself; it presupposes its resolution and inhabits one of the two resulting worlds.

6.2. Real closed fields and certified quantifier elimination.

The algebraic and model-theoretic backbone is the theory of real closed fields. Tarski’s quantifier elimination makes them the natural setting for definability questions involving addition, multiplication, and order [3, 4]; the standard real-algebraic reference is Bochnak–Coste–Roy [12].

On the formalization side, the gap `RCFModelComplete` has substantial prior art in other proof assistants. Cohen and Mahboubi formalized, in Coq/SSReflect, a certified quantifier-elimination procedure for the first-order theory of real closed fields, yielding decidability with a correctness proof [8] (the `math-comp/real-closed` library). McLaughlin and Harrison implemented a proof-producing decision procedure for RCF in HOL Light [9]. Neither development transfers directly to Mathlib’s deep-embedded `FirstOrder` framework, but Cohen–Mahboubi is the closest blueprint for the remaining gap; Section 5 explains why the cheaper categoricity route used for Mathlib’s ACF completeness is unavailable for RCF.

6.3. Formalized mathematics in Lean and Mathlib.

The project belongs to the growing ecosystem of Lean 4 and Mathlib formalizations [13, 14]. Two parts of Mathlib carry the present development: the model theory library (elementary substructures, Skolem functions and downward

Löwenheim–Skolem, `Language.DirectLimit`, and the deep-embedded algebra of `Language.ring` up to the completeness of ACF) and the cardinal/ordinal library (König’s theorem, cofinality, fundamental sequences). ICAH is not only a Lean artifact but also a diagnostic of Mathlib boundaries: it identifies concrete places where model theory, real closed fields, and directed colimits need stronger APIs — most pointedly, the absence of any elementarity result for `Language.DirectLimit` and of model completeness for RCF.

6.4. Elementary chains and direct limits.

Elementary-chain arguments are standard in model theory; the canonical reference is Chang–Keisler [5], and the König/cofinality background is in Kunen [15] and Jech [16]. The formal contribution here is more specific: it tests how far Mathlib’s `Language.DirectLimit` API can be pushed toward a fully formal elementary-chain theorem. The relativized elementarity theorem (`tarskiVaughtDirectLimit`, via `DirectLimit.lift` and the Tarski–Vaught test), the **ambient-free** chain theorem (`ofLevelElem`: the direct limit of an elementary chain is an elementary extension of each level, by formula induction), and the cardinality identity (`directLimit_card_eq_iSup`) are all proved locally; they are natural candidates for upstreaming, since Mathlib’s `DirectLimit` file currently has no elementarity results.

7. NEIGHBORING FIELDS AND ANALOGUES

The ICAH architecture instantiates a pattern — carrier, closure property, well-behaved first-order theory, elementary embeddings, direct limits, cardinal book-keeping — that has natural analogues beyond real closed fields. The closest is the p-adic side: a p-adically closed analogue would replace subfields of $\mathbb{R}$ by valued subfields of a p-adic ambient field, real closure by p-adic closure or henselianity, and the ring language by a valued-field language, with Macintyre’s definability theory playing the role of Tarski’s [17, 18]. Henselian valued fields offer a broader comparison class with the same shape, and algebraically closed fields a simpler control case (whose Mathlib model theory is, notably, already complete enough to serve as the template discussed in Section 5). These analogies are deliberately deferred to future work: the core contribution stays focused on the formal Lean development around the real continuum.

8. CONCLUSION, REPRODUCIBILITY, AND ENGAGEMENT PLAN

The current ICAH formalization is best presented as a rigorous formalization study rather than as a finished foundational theory. Its strongest components are already useful beyond ICAH itself: the DLS-based existence of elementary substrata of $\mathbb{R}$ at every intermediate cardinality (`exists_elementary_substratum`), the real-closedness of $\mathbb{R}$ (`Real.isRealClosed`), the existence of real-closed subfields of $\mathbb{R}$ of every infinite cardinality up to the continuum (`exists_rc_subfield`), the Tarski–Vaught elementarity theorems for direct limits — both relativized (`tarskiVaughtDirectLimit`) and ambient-free (`ofLevelElem`) — the direct-limit cardinality identity and closure theorem (`directLimit_card_eq_iSup`, `directLimit_card_lt_continuum`), and the $\text{cof}(c)$ sharpness pair for exhausting families.

The most important editorial decision is to keep the paper honest about the distinction between proved Lean results and named hypotheses. This is not a weakness: the hypothesis inventory — `NotCH` for the regime, `RCFModelComplete` for Pillar B only, both visible in type signatures, with the kernel-level audit machine-checked by `#guard_msgs` — turns an ambitious mathematical program into precise, independently attackable formalization problems. That the two sharpest findings of the latest review round (the vacuous cardinality theorem and the Löwenheim–Skolem shortcut) were visible **from the development’s own documentation** is evidence the methodology does what it claims.

8.1. Reproducibility.

The development is pinned and machine-checked:

- **Toolchain:** `leanprover/lean4:v4.31.0-rc2`.
- **Mathlib revision:** `a810615ff479602ad66b5403d179bfa805314a50` (June 2026), locked in `lake-manifest.json`.
- **Repository:** [19], with CI running `lake build` (which enforces the `#guard_msgs` audits), a zero-sorry check, and a zero-axiom check.
- **Statistics:** 11 Lean modules under `ICAH/`, circa 1,800 lines of Lean, 0 axioms, 0 sorries; both main theorems audit to [`propext`, `Classical.choice`, `Quot.sound`].

8.2. Declaration-to-paper mapping.

Lean declaration	Paper role	Hypotheses
<code>NotCH</code>	the $\neg \text{CH}$ regime (Def., Section 5)	—
<code>RCFModelComplete</code>	RCF model completeness (Gap, Section 5)	—
<code>exists_intermediate_cardinal</code>	intermediate cardinal witness	<code>NotCH</code>
<code>Stratum</code> , <code>SizeAwareField</code>	core objects (Section 3)	—
<code>syntheticStratum</code>	concrete $\aleph_1$ -stratum	<code>NotCH</code>
<code>exists_elementary_substratum</code>	Pillar A extending via DLS	ZFC-pure

Lean declaration	Paper role	Hypotheses
<code>elemSubstratumSubfield</code>	substrata are subfields	ZFC-pure
<code>exists_rc_subfield</code>	Pillar B strata, every $\aleph_0 \leq \kappa \leq c$	ZFC-pure
<code>relAlgebraic_isRealClosed</code>	relative algebraic closures are RC	ZFC-pure
<code>Real.isRealClosed</code>	$\mathbb{R}$ is real closed	ZFC-pure
<code>fieldOnStratum</code>	packaging lemma (Section 4)	ZFC-pure
<code>tarskiVaughtDirectLimit</code>	chain theorem rel. $\mathbb{R}$	ZFC-pure
<code>ofLevelElem</code> / <code>realize_ofLevel_iff</code>	ambient-free chain theorem	ZFC-pure
<code>directLimit_card_eq_iSup</code>	limit cardinality identity	ZFC-pure
<code>directLimit_card_lt_countable</code>	closure under countable chains	ZFC-pure
<code>cofinal_family_length_lower_bound</code>	club bounds, lower bound	ZFC-pure
<code>exists_cofinal_rc_family</code> (cofinal length)	(cofinal length) lies, optimal length	NotCH
<code>cofinal_family_limit_size</code>	M6 limit-size clause	NotCH
<code>ICAHElementary</code> / <code>icahElementary</code>	Theorem 1 (Pillar A)	NotCH
<code>ICAHStatement</code> / <code>icahTheorem</code>	Theorem 2 (Pillar B)	NotCH, <code>RCFModelComplete</code>

8.3. Engagement plan.

1. Post on the Lean Zulip model-theory stream with the repository and a minimized in-framework statement of `RCFModelComplete`; this may surface in-progress RCF or real-closure work and prevents duplicated upstreaming. (A draft is maintained in `docs/UPSTREAMING.md`.)
2. PR sequence: (1) DLS-based elementary substrata of $\mathbb{R}$ plus `directLimit_card_eq_iSup` — small, ZFC-pure, attractive; (2) the ambient-free Tarski–Vaught chain theorem for `Language.DirectLimit`; (3) `Real.isRealClosed`, `isRealClosed_of_forall_root`, `relAlgebraic_isRealClosed` after the dedup check; (4) long-term, RCF model completeness modeled on the ACF files via Robinson’s test.
3. Venue: this is a formalization paper — ITP, CPP, or the Journal of Automated Reasoning, with an arXiv preprint cross-listed math.LO/cs.LO. It should not be sent to a set-theory venue as-is, where the mathematics reads as exercises.

4. Keep p-adic and valued-field analogies as future work, not as the main thesis.

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